

FACULTY OF COMPUTING UNIVERSITI TEKNOLOGI MALAYSIA

DATA MINING (SECP2753) SUPERVISED LEARNING (CLASSIFICATION)

CLASS ACTIVITY (CLASSIFICATION)

Q1 - What is classification in data mining?

Q2 – Explain the importance and real-world applications of classification in data mining.

Q3 – Identify **at least TEN (10)** classification algorithms. Then, provide a concise explanation (including an example) for **THREE (3)** of them.

Q4 – How does classification works? Describe the general steps involved in building a classification model.

Q5 – Identify and explain common evaluation metrics used in classification in data mining (e.g. accuracy, precision, recall, F1-score). Include relevant formulas.

Q6 - Compare Supervised Learning and Unsupervised Learning with examples.

Q7 – Differentiate between binary classification and multi-class classification. Provide an example for each.

Q8 – Discuss common challenges encountered in classification tasks.

CLASS ACTIVITY (CLASSIFICATION)

- Q9 - What is a **confusion matrix** and how is it used to evaluate classification models? Provide a labeled example.
- Q10 – Explain the concept of **overfitting** and **underfitting** in classification models. How can each issue be addressed?
- Q11 – Describe the role of **cross-validation** in evaluating the performance of a classification algorithm.
- Q12 – How can **feature selection** improve classification model performance? Mention at least two methods.
- Q13 – Compare **decision trees** and **logistic regression** in terms of model interpretability, accuracy, and use cases.
- Q14- Explain what **ensemble methods** (e.g., Bagging, Boosting, Random Forest) are and how they improve classification performance.